DIFFRACTION 4

Resnick, Halliday, Krane

Q.1 What - is diffraction. Explain single slit diffraction and derive diffraction equation? (sU. 2011, 2014).

DIFFRACTION

The slight bending of lights when it passes around the edge of an object is called diffraction.

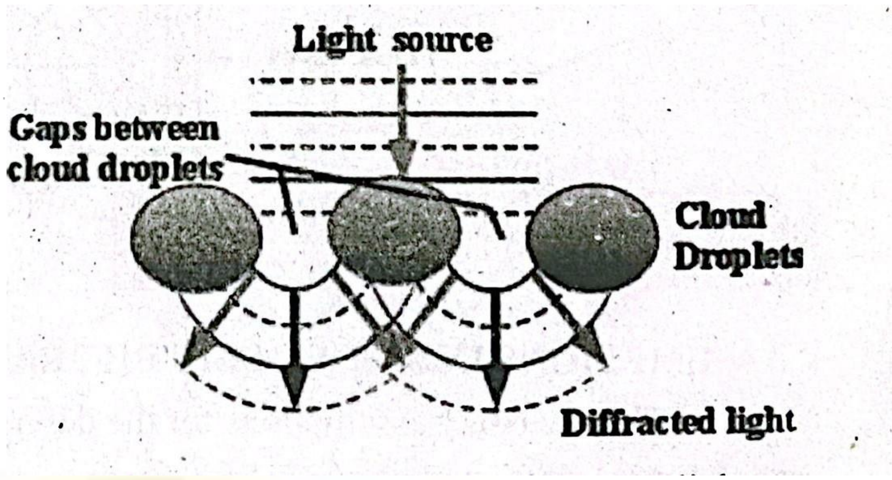
The amount of bending depends on the relative size of the wavelength of light to the size of the opening. The bending of light will be almost unnoticeable when opening is much larger than wavelength of light. The amount of bending IS considerable when both are comparable in size or equal. The bending can be seen easily with the naked eye.

The phenomenon• of diffraction is used to distinguish waves from particles

because wave diffracts and particles do not diffract.

EXAMPLE

The tiny water droplets found in clouds are called atmospheric particles. The silver lining found around the edges of clouds or coronas surrounding the Sun or moon is result of diffraction.



The fig. below shows how light from Sun or moon is bent around small droplets in the cloud.

The light waves are altered and interact with one another when sunlight or moonlight encounters a cloud droplet. The light will appear brighter when there is

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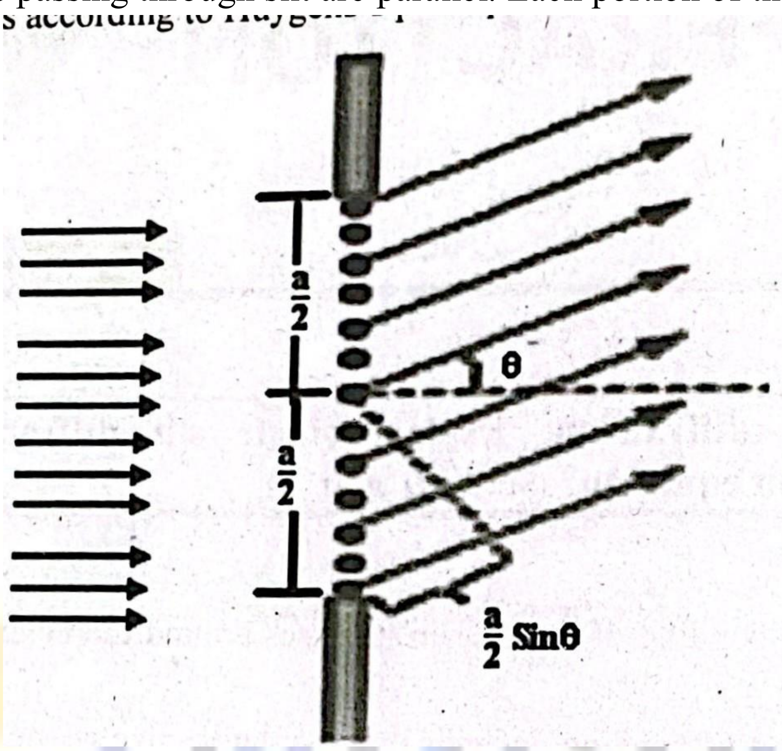
[M.B.DI WAVES & OSCILLATIONS For B.Sc. Classes Based on Resn: constructive interference. The light will appear darker or disappear entirely when there is destructive interference.

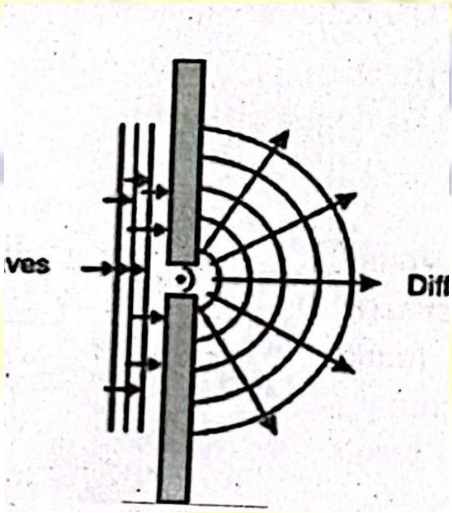
SINGLE SLIT DIFFRACTION

Consider a plane wave is falling normally on a single slit having width a as shown in fig. All rays passing through slit are parallel. Each portion of the slit will act as a source of



light waves according to Huygens's principle.

The diffraction pattern of the resulting wave can be calculated by treating each point in the aperture as a point source from which new waves spread out when light encounters the slit.

Plano  waves

Diffracted waves

ASSUMPTIONS IN SINGLE SLIT DIFFRACTION

The necessary assumptions for the description of single slit diffraction pattern are O The slit size must be small relative to the wavelength of light.

@ The screen must be very far away.

O The intensity of light at any point on the screen is independent of the angle

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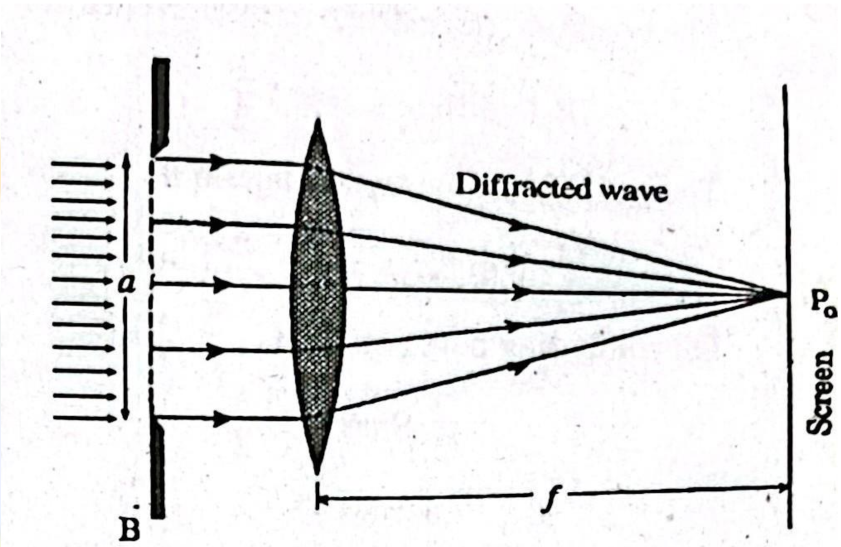
made between the ray to the screen and the normal line yetween the slit and

cENTRAL MAXIMA IN SINGLE SLIT DIFFRACTION

Consider a central point PO on the screen. The rays that leave the slit are parallel to the central horizontal axis and are brought to a focus at PO by convex lens. These rays are in phase at the plane of the slit and remain in phase when are brought to focus by the lens at PO. These rays are in phase and interfere constructively. The fesult is maximum intensity at PO called central maxima.

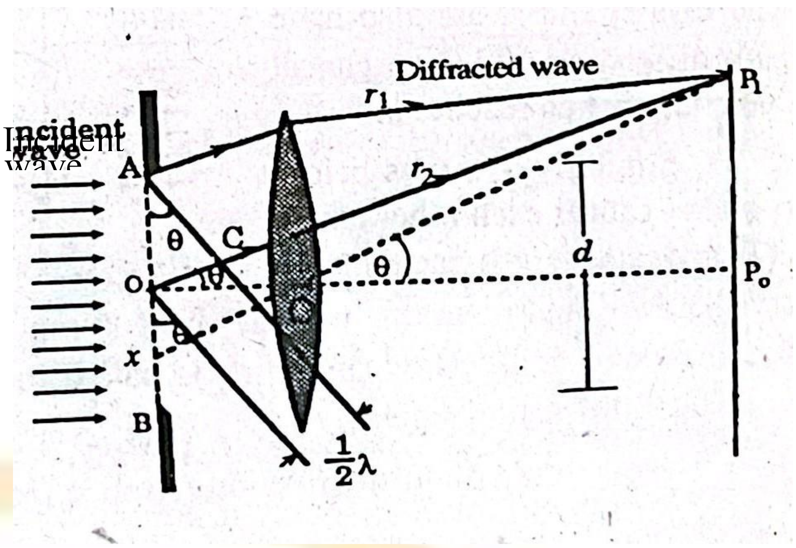


FIRST MINIMA IN SINGLE SLIT DIFFRACTION

Now consider a point P f on the screen for the formation of first minima of diffraction pattern. Divide the slit into two halves. Each ray from upper half will be 1800 out of phase with corresponding ray from lower half. For example suppose there are 10 point sources. The first 5 are in lower half and 6 to 10 are in upper half. Source I and source 6 are Separated by distance a/2. Similar observation applies to source 2 and source 7 and any pair having separation a/2.

The light rays reaching at PI after leaving the slit makes angle 0 with central horizontal line as shown in fig.

The ray XP I ^A^ passes un— deflected through the center of lens and determines angle 0.



The ray rl coming from point source at the top of the slit and ray r2 coming from point source at center of slit

meet at Pl. The ray rl and r2 interfere destructively at PI if path difference is equal to 1/2 because both rays are out of phase. The path difference OC can be made

equal to 1/2 by proper selection Of O..

The same is true for a ray just below rp and another just below r2. Eyery ray arriving at PI from the upper half of the slit interferes destructively with one coming from the bottom half qf the slit. The intensity at Pi is zero. Now point Pi is called first minimum of diffraction pattern.

The condition of destructive interference for first minima is

Path difference — _

The path difference from diagram is

a

Path difference = Sino

On comparing both equations

a

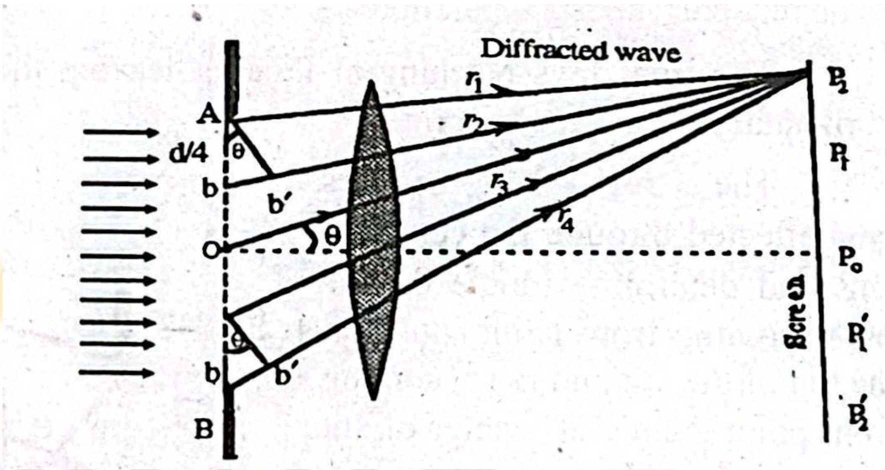
— 2 Sino 5

a Sino= λ

This shows central maxima becomes wider when slit is made narrower. The first minima occurs at $0 = 90^0$ if a = X. It means central maximum fills the entire forward hemisphere.

SECOND MINIMA IN SINGLE SLIT DIFFRACTION

Consider a point P2 on the screen for the formation of second minima of diffraction pattern. Divide the slit into four equal zones such that dray is leaving from the i top of each zone as shown in fig.



The value of 0 is so chosen that path difference bb' =V2. The rays rt and r2 cancel each other when reaches at P2. The rays r3 and r4 are also have path difference of 1/2 so cancel each other when reaches at P2.

Similarly two rays below rl and r2 cancel each other effect at P2 and sme is also frue for two rays below r3 and r4. In this way all rays coming from slit cancel each other and intensity becomes zero at P2. Now point P2 is called second point of. zero intensity or second

minima of diffraction pattern.

The condition of destructive interference for second minima is

Path difference — _

2

The path difference from diagram is

Path difference = a Sino

On comparing both equations

a

, Sino= –

a Sino= 2 λ

This is position of second minima of diffraction pattern on the screen.

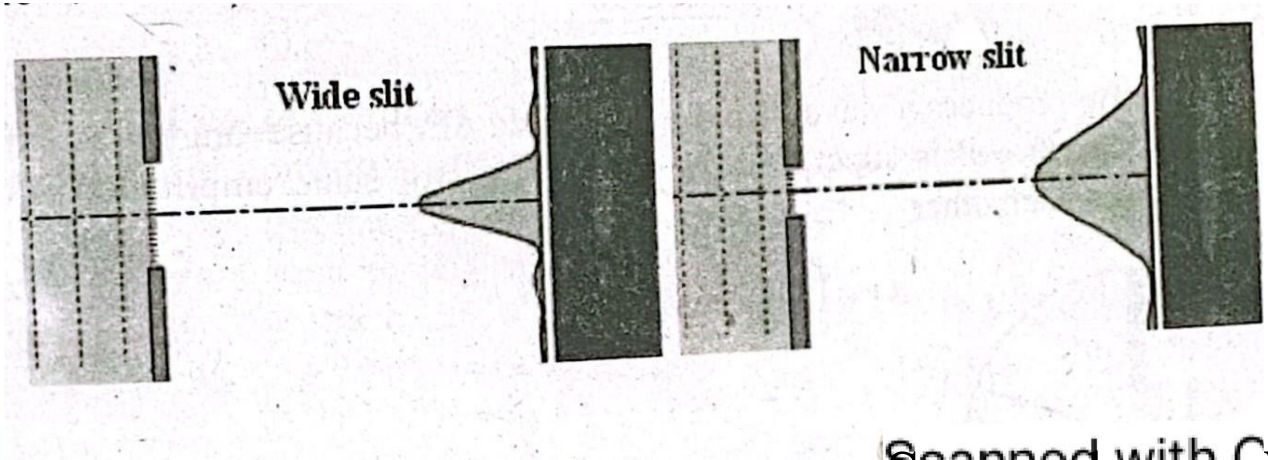
GENERALIZATION

The general formula for minima in the diffraction pattern on screen is

aSin0 = m X

Where m = ±l, ±2, ±3, ±4, ± 5------

This is called diffraction equation. The narrower slit will give a wider diffraction pattern while wider slit will give a narrower diffraction Pattern as shown in fig. below.


Wide slit
Narrow slit

POLARIZATION 5

Resnick, Halliday, Krane

Q.l What is unpolarized and polarized light, describe polarization of light briefly? (PU. 2003, 201 1)

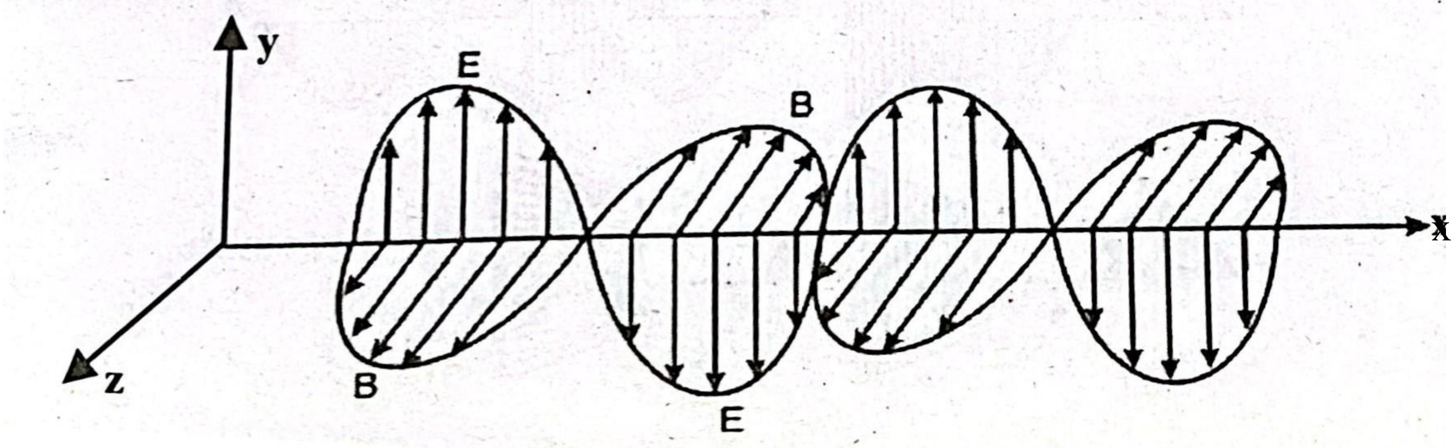
UNPOLARIZED LIGHT

The ordinary white light is made up of waves that fluctuates at all possible angles and vibrating in more than one plane is called unpolarized light. The light emitted by the sun or candle flame is unpolarized light. Such electromagnetic light waves are created due to vibrations of electric charges in a variety of directions.

POLARIZATION

The interference and diffraction have confirmed the wave nature of light but it is not decided through these phenomena whether light waves are transverse or longitudinal. The phenomenon which decided the transverse nature of light is called polarization.

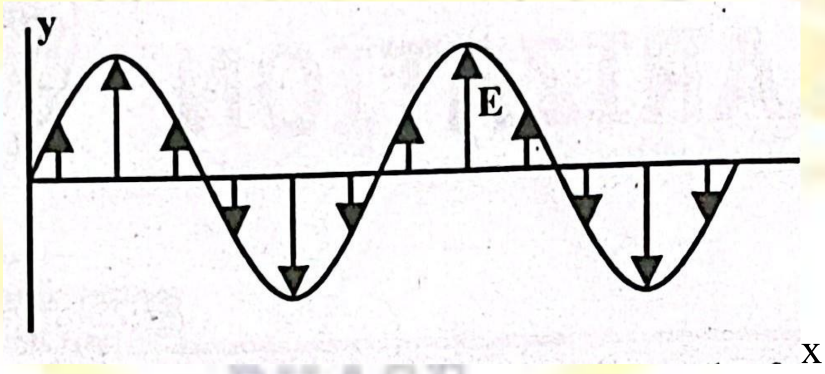
The light is a transverse electromagnetic waves made up of mutually perpendicular fluctuating electric field vectors and magnetic•field vectors. The diagram below shows that electric field vectors are in xy plane, magnetic field vectors are in xz plane while wave propagates along x direction.



wave propagates 512 [M.B.D] WAVES & OSCILLATIONS For B.Sc. Classes Based on Resnick, Halliday,

DIRECTION OF POLARIZATION OF LIGHT

The direction of polarization of light waves refers to the direction of E vectors of the waves. The diagram below shows a line traced out when electric field vectors propagate. Traditionally, only the electric field vectors are considered because magnetic field component is essentially the same.



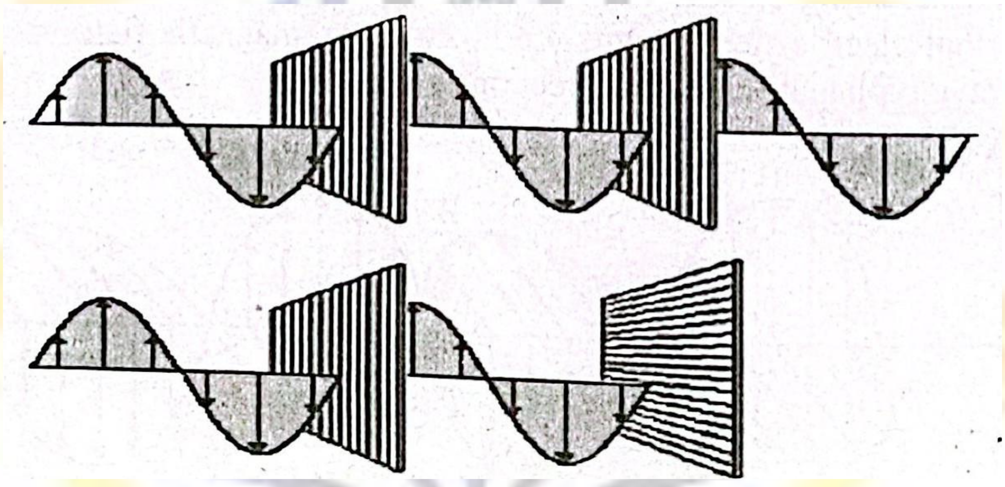
This time varying electric field can be thought of as a length of a rope held by two children at opposite ends. The ~~transverse waves are~~ produced when children begin to displace the ends of a rope in •such a way that it moves in a plane-up and down, left and right or at any angle in between them.

POLARIZER

The material which polarizes the light is called polarizer. It is an optical filter that allows to pass light of a specific polarization and blocks waves of other polarizations. It converts mixed polarized beam of light into a well defined polarized beam of light.

The polarizers are called linear polarizers and circular polarizers. Polarizers are used in many optical techniques and instruments. The polarizing fifiéös¯flöd applications in photography anUiquid crystal display technology.

The light passes through both polarizers when they are set up in series in such a way that their optical axes are parallel. The polarized light from the first is extinguished by the second when optical axes are set up, 90^0 apart or crossed. The amount Of transmitted light decreases when polarizer is rotated through 0^0 to 90^0 as demonstrated in the diagram below.

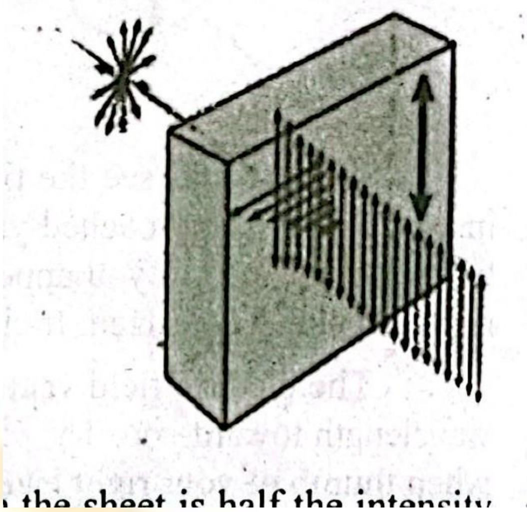


Q.5 What arc polarizing sheets? Write lhcir characteristics briefly. (PU. 2009)

POLARIZING SHEET

The commercial polarizing material is called Polaroid or polarizing shcct. The common types of polarizers are called linear polarized and circular polarized. The polarizer sheet converts a mixed polarized beam of light into a well defined polarized beam of light.

The polarizing sheet are used in many optical techniques and instruments. The polarizing filters find applications in photography and liquid crystal display technology.

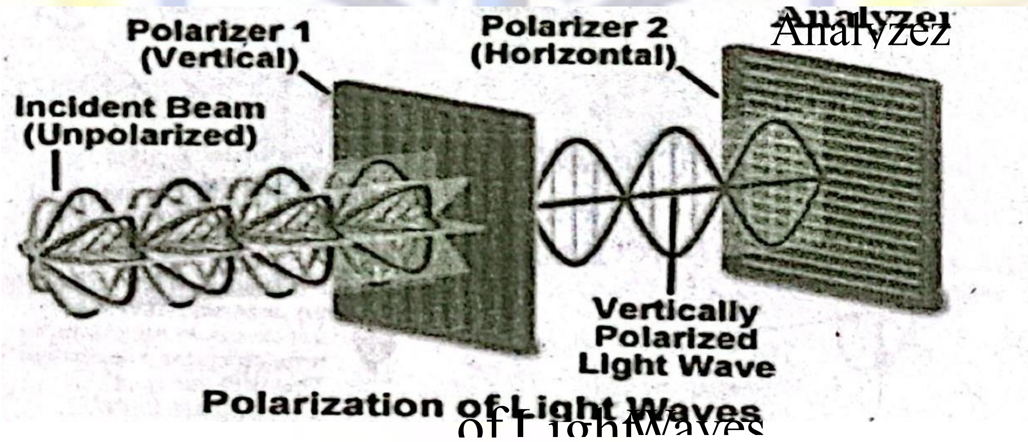


CHARACTERISTICS POLARIZING SHEET

The polarizing sheet has certain polarizing directions as indicated by parallel lines. These parallel lines allow to transmit only those wave train components whose electric field vectors vibrate parallel to this direction and absorbs those that vibrate at right angles to this direction. The light emerging from the sheet is called linearly polarized. The polarizing direction of sheet is established during manufacturmg process.

The intensity of the polarized light emerged through the sheet is half the intensity of the incident light when un-polarized light is incident on an ideal polarizing sheet. The orientation of the sheet does not count. The real polarizing sheets may transmit only 40% of the incident intensity because of reflection and partial absorption of the light along the polarizing direction.

Consider a polarizing sheet called Polaroid and another polarizing sheet called analyzer whose polarizing directions are perpendicular to one another. The Polaroid and analyzer are placed in front Of un-polarized light as shown in fig.



The intensity of polaroid light falls on screen placed next to analyzer •whe^{n} polarizing direction of polaroid and analyzer are parallel. The intensity on scree^{n}

disappears when analyzer is rotated through 900. Thus phenomenon of polarization proves that light is transverse in nature.

Assume that amplitude of linearly polarized light incident on analyzer is Eo. The

amplitude of light emerges through analyzer is Eo Cos0. Where 0 is the angle between the polarizing directions of polaroid and analyzer. The intensity of light beam is proportional to square of amplitude. The transmitted intensity of light is given by Malus law.